

Tangents and normals



1

a

i $y = 2x + 2$

ii $y = -\frac{x}{2} + 2$

b

i $y = 5x + 5$

ii $y = -\frac{x}{5} + 5$

c

i $y = \frac{x}{6} + \frac{1}{6}$

ii $y = -6x + \frac{1}{6}$

d

i $y = 10x + 4$

ii $y = -\frac{x}{10} + 4$

2

a

i $\frac{e}{2}$

ii $ex - 2y + e - \frac{2}{e} = 0$

b

i $-\frac{1}{2e}$

i $x + 2ey - 2e^2 - 1 = 0$

c

i $-\frac{e^2}{e^2 + 1}$

i $e^2x + (e^2 + 1)y - 4e^2 + \frac{1}{e^2} - 1 = 0$

3 $k = 3$

4

a $y' = (1 - x)e^{-x}$

b $\frac{e^3}{2}$

c $y = \frac{e^3}{2}(x - 3) + 3e^{-3}$

5

a $3e$

b $y = 3ex - 2e$

c $A = \left(\frac{2}{3}, 0\right)$

d $B = (0, -2e)$

e $\frac{2e}{3} \text{ units}^2$

6

a $(0, 3)$

b -3

c 72°

d -1

e 45°

f 27°

7

a 81.9°

b 81.9°



c 16.2°

8 No solution provided.

Exponential graphs

9

a $f(0) = 2$

b $f'(0) = 1$

c Increasing, because $f'(x) > 0$ for all real x .

d 3

10

a $f(0) = \frac{4}{5}$

b $f'(t) = \frac{12}{e^t(2 + 3e^{-t})^2}$

c Increasing, because $f'(t) > 0$ for all real t

d 0

e 2

11

a $(2, -e^2)$

b Minimum

12

a $f'(x) = -8xe^{-x^2}$

b

i $x = 0$

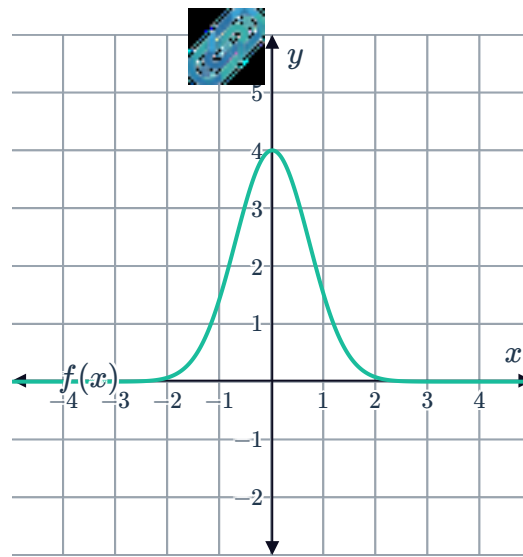
ii $x < 0$

iii $x > 0$

c 0

d 0

e



13

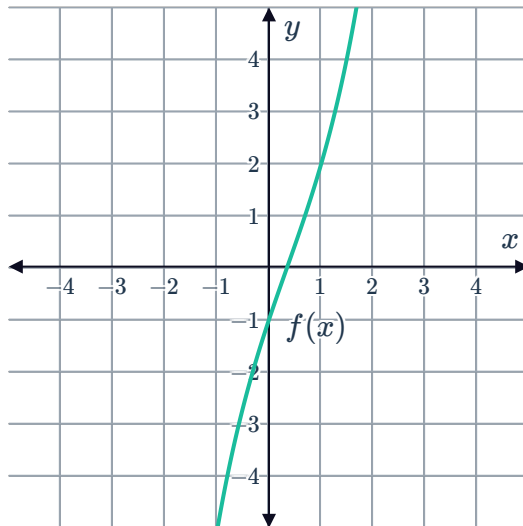
a $x = 0.35$

b $y = -1$

c $y \rightarrow \infty$

d $y \rightarrow -\infty$

e



14

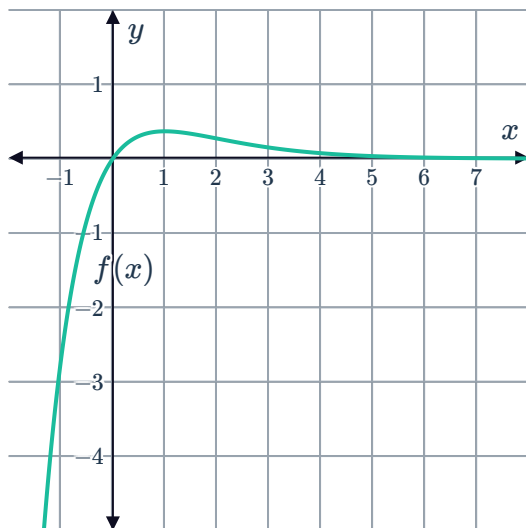
a

i $x = 0$

iii $\left(1, \frac{1}{e}\right)$

v $\left(2, \frac{2}{e^2}\right)$

vii



ii $y = 0$

iv Maximum stationary point

vi As $x \rightarrow \infty, y \rightarrow 0$.

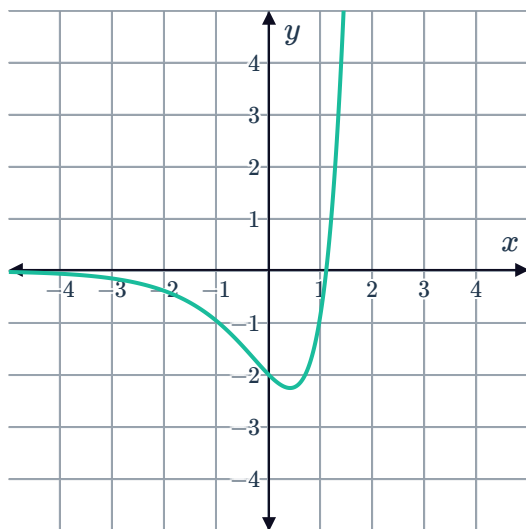
b

i $x = \ln 3$

iii $\left(\ln \frac{3}{2}, -\frac{9}{4}\right)$

v $\left(\ln \frac{3}{4}, -\frac{27}{16}\right)$

vii



ii $y = -2$

iv Minimum stationary point

vi As $x \rightarrow \infty, y \rightarrow \infty$.

15

a $x = 0$

b $y = 0$

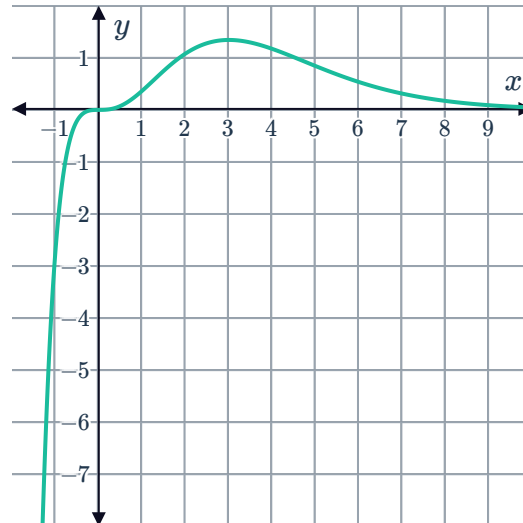
c $y' = 3x^2e^{-x} - x^3e^{-x}$

d $(0, 0), (3, 1.34)$



e Point of inflection at $x = 0$ and maximum at $x = 3$.

f



16

a $x = 0$

c $y' = \frac{e^{-x}}{2\sqrt{x}} - e^{-x}\sqrt{x}$

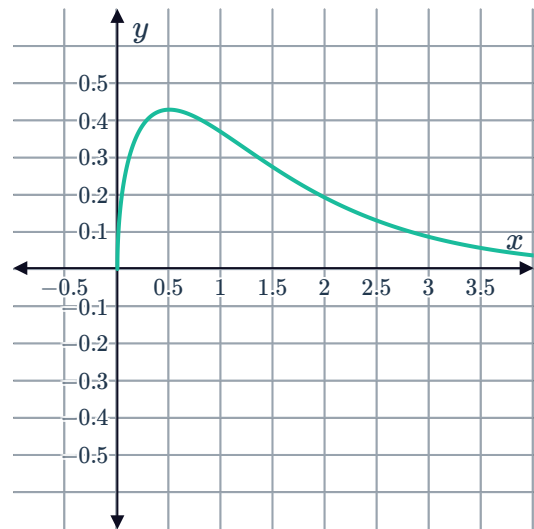
e Maximum



b $y = 0$

d $(0.50, 0.43)$

f



17

a $x = \ln 3$

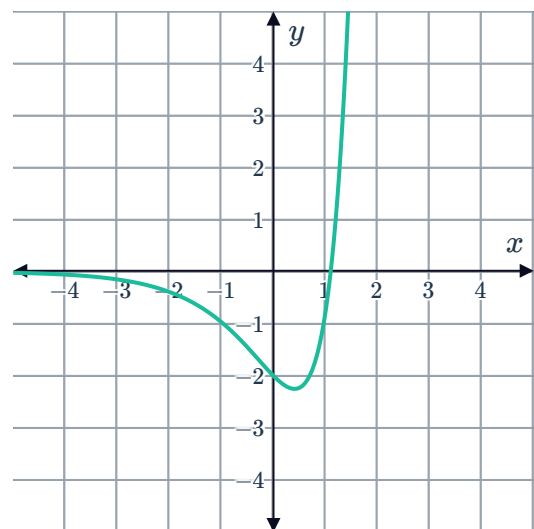
c $(0.41, -2.25)$

e $x \rightarrow \infty, y \rightarrow \infty$
 $x \rightarrow -\infty, y \rightarrow 0$

b $y = -2$

d Minimum stationary point

f



Exponential growth and decay

18

a $y = 840(0.96)^m$

b 606 rodents

19

a 8 million



b

i 7130 million

ii 9286 million

20

a $t = 29$ years

b $t = 22.3$ years

21

a 1105 km^2

b 332 km^2

22 5.08%

23

a 0

b $t = \frac{1}{4} \text{ hr}$

c 0.736

d No

24

a Metal M

b The parameter k .

25

a 1035° C

b 25° C

26

a $A = -5, B = -170$

b $t = 73$ minutes

c $\frac{dT}{dt} = -0.6$

27

a $f(t) = 1.27(0.92)^t$

b 8%

c \$0.47